

Day 5 – Model Evaluation Metrics (Classification & Regression)

This is a **content-first Day 5 teaching presentation**.

The goal of Day 5 is to make participants **comfortable with evaluation metrics conceptually and manually**, not just via libraries.

This session focuses on: - Understanding **why evaluation metrics matter** - Knowing **which metric to use for which problem** - Being able to **calculate and explain metrics by hand** (interview-critical)

Slide 1 — Title

Day 5: Model Evaluation Metrics

How we measure whether a model is actually good

Slide 2 — Why Model Evaluation Matters

Training a model is easy.

Knowing whether it is *good* is hard.

Evaluation answers: - Is the model useful? - Is it better than a baseline? - Does it solve the business problem?

A model without evaluation is not engineering — it's guessing.

Slide 3 — Recap: Problem Type Drives Metrics

From earlier sessions: - Classification → predicts categories - Regression → predicts numbers

Important rule:

Metrics depend on the *problem type*, not the algorithm.

Slide 4 — Two Worlds of Evaluation

There are two major evaluation categories:

1. **Classification Metrics**
2. **Regression Metrics**

We will cover both — conceptually and manually.

PART 1 — CLASSIFICATION METRICS

Slide 5 — Confusion Matrix (Foundation)

A confusion matrix shows: - Actual vs Predicted outcomes

Four outcomes: - True Positive (TP) - False Positive (FP) - True Negative (TN) - False Negative (FN)

This matrix is the foundation of all classification metrics.

Slide 6 — Confusion Matrix (Example)

Example: Spam Detection

Actual \ Predicted - Spam → Spam = TP - Spam → Not Spam = FN - Not Spam → Spam = FP - Not Spam → Not Spam = TN

Every classification metric comes from these four numbers.

Slide 7 — Accuracy

Accuracy = (TP + TN) / Total

Meaning: - Percentage of correct predictions

Problem: - Misleading for imbalanced datasets

Example: - 95% accuracy but misses all fraud cases

Slide 8 — Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Meaning: - Of all predicted positives, how many were correct?

Use when: - False positives are costly

Example: - Spam incorrectly marked as important email

Slide 9 — Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Meaning: - Of all actual positives, how many did we catch?

Use when: - Missing positives is costly

Example: - Fraud or disease detection

Slide 10 — Precision vs Recall Trade-off

High precision → fewer false positives

High recall → fewer false negatives

You usually cannot maximize both.

Business goals decide which one matters more.

Slide 11 — F1 Score

F1 Score = Harmonic mean of Precision & Recall

Used when: - Need balance between precision and recall - Dataset is imbalanced

Common interview metric.

Slide 12 — Manual Exercise (Classification)

Given: - TP = 40 - FP = 10 - FN = 20 - TN = 30

Calculate by hand: - Accuracy - Precision - Recall - F1 Score

This is **mandatory practice**.

PART 2 — REGRESSION METRICS

Slide 13 — Why Regression Metrics Are Different

Regression predicts **numbers**, not labels.

We measure: - How far predictions are from actual values

Errors matter more than exact matches.

Slide 14 — Mean Absolute Error (MAE)

MAE = Average(|Actual - Predicted|)

Meaning: - Average error magnitude

Pros: - Easy to interpret - Less sensitive to outliers

Slide 15 — Mean Squared Error (MSE) & RMSE

MSE: Average squared error

RMSE: Square root of MSE

Meaning: - Penalizes large errors more heavily

RMSE is commonly used in practice.

Slide 16 — R^2 (R-Squared)

R^2 explains: - How much variance in the target is explained by the model

Range: - 0 to 1 (sometimes negative)

Important: - High R^2 does not guarantee a good model

Slide 17 — MAPE (Mean Absolute Percentage Error)

MAPE = Average(|(Actual – Predicted) / Actual|)

Meaning: - Error expressed as a percentage

Use when: - Relative error matters more than absolute

Avoid when: - Actual values can be zero

Slide 18 — Manual Exercise (Regression)

Actual: [100, 200, 300] Predicted: [90, 210, 290]

Calculate by hand: - MAE - RMSE - MAPE

Focus on *interpretation*, not formulas.

Slide 19 — Choosing the Right Metric

Guidelines: - Classification + imbalance → Precision / Recall / F1 - Regression + business cost → MAE or RMSE - Percentage sensitivity → MAPE

Metrics should align with business impact.

Slide 20 — Common Mistakes

- Using accuracy for imbalanced data
 - Optimizing the wrong metric
 - Comparing models using different metrics
 - Ignoring business meaning of errors
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Slide 21 — Interview Perspective

You should be able to answer: - Why not accuracy? - When is recall more important? - Difference between RMSE and MAE? - What does R^2 actually mean?

This session directly improves interview confidence.

Slide 22 — Key Takeaways

- Metrics define success, not models
 - Confusion matrix is the foundation
 - Always align metrics with business goals
 - You must be able to explain metrics without code
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Slide 23 — Homework / Practice

- Recalculate metrics from your Day 3–4 model
 - Explain which metric matters most and why
 - Be ready to justify this choice verbally
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End of Day 5 Teaching Presentation